

ROOF-TOP SOLAR PANELS FOR HOMES AND FACTORIES

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In today's world, renewable energy plays an important role in the development and advancement of several industries. The conventional sources of energy, such as coal and petroleum, are almost coming to an end, and the world would be without natural resources by the end of 2050, claim experts. Hence, we need to find an alternative to save these resources and the world from their harmful impact on our lives.

Solar energy is an impeccable example of a renewable energy whose popularity has been increasing all over the world. It is an important natural source of energy besides hydropower and wind energy. It will play a pivotal role in protecting the future of our planet. Harnessing solar energy by installing solar panels can make a significant difference to the planet. In addition, solar panels can be installed easily.

There are two types of solar panels available for your home or factory—Grid-tied solar power systems without batteries and off-grid solar power systems. It makes even more sense to install roof-top solar panels in factories. Reason being, factories generally work in day time, when the sunlight is available in abundance. So battery back-up need not be very powerful in case of even stand-alone off-grid systems.

Before finding out the latest rooftop panels available in the Indian market, let's find out the

essential criteria that one must look for before buying these:

Don't make your choice based on advertisements. As the popularity of solar panels is increasing, many companies are stepping into this business, and they are trying hard to influence people with their adverts. But before buying, you must ask your close associates, relatives, or neighbours who have installed solar kits in their homes about the brand or the company they have selected. Most importantly, you need to ask about three things: service quality, ease of installation, and problems, if any.

Understand the material used. While buying the off-grid solar power systems, the first thing that you need to be careful about is the type of panel that you are selecting. There are two kinds of panels available in the market:

Monocrystalline. These panels are crafted with single-crystal silicon bars that are cut into wafers. Monocrystalline panels have a black hue and are more efficient, thereby requiring lesser space. These panels can absorb massive amounts of solar energy in strong sunlight. But absorption and thus their output is reduced when light reduces.

Polycrystalline. These are purified and moulded crystal panels having a bluish hue. These are less efficient but cheaper than the monocrystalline panels.

Both of these panels have their unique properties. You must choose the one that fits your needs and budget.

Think about budget. The most important aspect is the budget before buying any product. India is a price-conscious market where people look for better quality products at lower prices. However, the solar kits for home are quite expensive. But if you want to get rid of frequent maintenance and repair

costs, you need to purchase the best panels available in the market. Also, electricity bills will decrease massively with the use of high-quality solar panels.

Know about location. Location plays an important role while selecting the solar panels. If you live in an area where there is uninterrupted strong sunlight most of the time of the year, you can opt for monocrystalline systems. But if you live in an area where direct or strong sunlight is scarce, go for polycrystalline systems. These will absorb sunlight from different directions.

Now, let's check out some solar panels available in the Indian market. Generally, the size of a roof-top solar panel is 165cm × 99cm (65 × 39 inches):

Panasonic 340N HIT. Panasonic has launched this 96-cell high-efficiency module for installers that support



projects with a higher voltage. Featuring high-efficiency n-type crystalline combined with amorphous layers, the 340N HIT modules provide better performance in heat and low light conditions. The new 340N HIT module also features a smaller footprint than competing products while providing the maximum energy yield and flexibility in constrained rooftop spaces.

Price: ₹ On request

Microtek 150W/12V polycrystalline panel. It gives optimal output with a good low-light response and